

Simulation 3D Fireworks Based on Particles System

Course: Computer Graphics

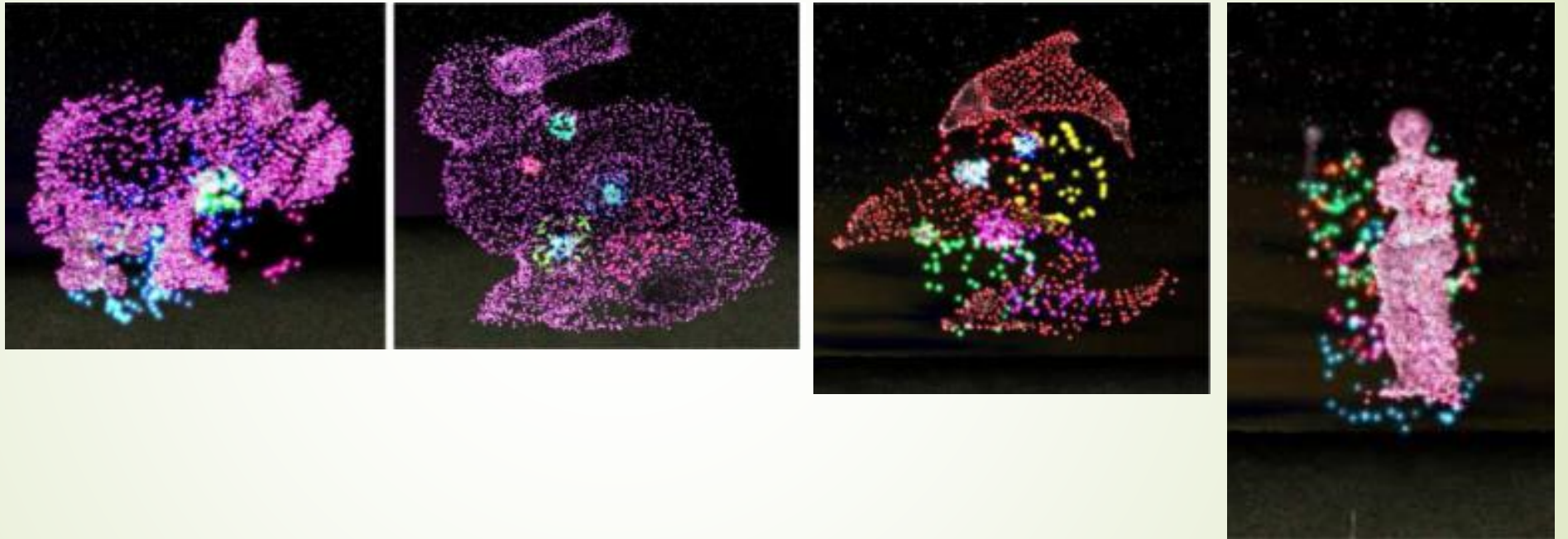
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Zhao, H., Fan, R., Wang, C. C. L., Jin, X., & Meng, Y. (2009). Fireworks controller.

Goal: construct a shape with particles after a fire explosion



Instance Mesh Algorithm

- ▶ Each particle is a little quad, it can be a sphere or something.
- ▶ call `glDrawArrays` once for each particle → very bad idea, all multiprocessors use to just draw a single quad.
- ▶ base mesh (in our case, a simple quad of 2 triangles), but many instances of this quad (10000).
- ▶ buffers: for the vertices of the mesh, particles' centers, particles' colors.
- ▶ use `glDrawArraysInstanced` / `glDrawElementsInstanced`, which is equivalent to calling `glDrawArrays` N times. But works parallel for buffers.

Explosion

- Random direction for each axis.
- Random value for initialize velocity, shouldn't be so small.
- (Tutorial not Paper)



Physics Formulas after the explosion

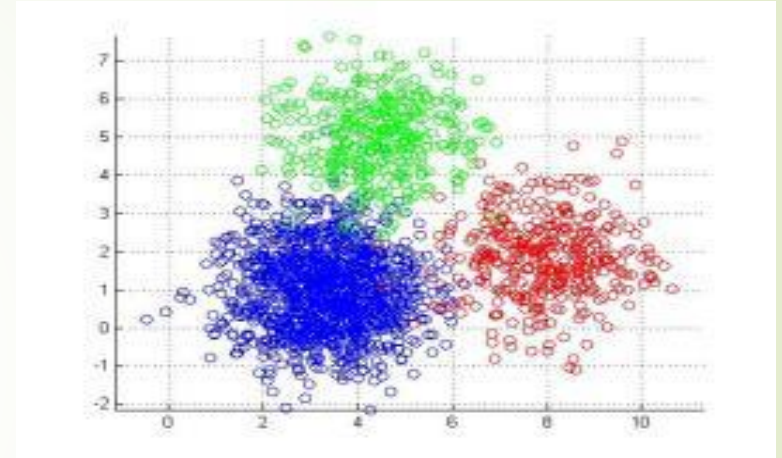
- M: initial mass of a particle,
 g: gravity acceleration,
 T: lifetime of the particle,
 T₀: time it takes to reach the cluster,
 t: time spend after explosion,
 k: air friction coefficient,
 S: distance of the point at t=0 with the cluster
- X,Z axis: (different S on each, but same formula)

$$C = S \frac{\left(\frac{kT}{M} + 1\right)}{T^{\frac{kT}{M}+1} - (T - T_0)^{\frac{kT}{M}+1}}$$

$$v = C(T - t)^{\frac{kT}{M}}$$

Sampling

- Clustering: kmeans, iterative method for finding center of point groups.
- Implemented in the application based on input mesh vertices.
- Hyperparameters: number of clusters, iteration



Physics Formulas after the explosion

- M: initial mass of a particle,
 g: gravity acceleration,
 T: lifetime of the particle,
 T₀: time it takes to reach the cluster,
 t: time spend after explosion,
 k: air friction coefficient,
 S: distance of the point at t=0 with the cluster
- Y axis (k issue in the paper, so ignored in implementation):

$$C = \left(S - \frac{Mg(2TT_0 - T_0^2)}{2(kT - M)} \right) \frac{\frac{kT}{M} + 1}{T^{\frac{kT}{M}+1} - (T - T_0)^{\frac{kT}{M}+1}}$$

$$v = C(T - t)^{\frac{kT}{M}} + \frac{Mg}{kT - M}(T - t)$$

References

- ▶ Zhao, H., Fan, R., Wang, C. C. L., Jin, X., & Meng, Y. (2009). Fireworks controller. *Computer Animation and Virtual Worlds*, 20, 185-194. <https://doi.org/10.1002/cav.287>
- ▶ <https://www.opengl-tutorial.org/intermediate-tutorials/billboards-particles/particles-instancing/>
- ▶ Hall, J. D., & Hart, J. C. (2004). GPU Acceleration of Iterative Clustering. *University of Illinois at Urbana-Champaign*.
- ▶ <https://www.istockphoto.com/photos/large-firework-explosion>
- ▶ <https://prateekvjoshi.com/2013/06/06/what-is-k-means-clustering/>

Result Videos attached.

Thanks for listening!

Any questions?